Template for MIE 402 Report (title)

Your name

(Group members' names in alphabetic order of their last name)

Abstract

Discuss briefly the main reason for this lab. What has been done, the theory that has been validated, and the main findings of your experiments. Be brief and to the point. This should be a **maximum of 10 lines**.

1. Introduction

Introduce the problem to someone who has some relevent background (assume the reader has passed Dynamics and Fluid Mechanics undergraduate courses). Give the theoretical background. Give the equation that governs the system. Discuss what you will be validating in the experiments that you will discuss in this report. (**1 page**)

2. Method

Discuss what you did in the lab. Focus on the important steps that you had to take to conduct the experiment. Discuss the measurement methods and tools. (**1 page**)

3. Results

Present results clearly and objectively. Focus on observable facts—discuss what you actually see, not what you expected or hoped to see (i.e., it should not be the simulation results unless the lab is a virtual/simulation-based lab). Use plots to show the system's behavior; figures are critical and should be prepared carefully. Ensure all text, lines, axis labels, tick values, and units are clearly legible (avoid tiny fonts or overly thin lines). Invest time to make figures clean and professional.

When generating figures in MATLAB, use Edit → Copy Figure (or insert vector formats such as .eps or .pdf on a Mac) and paste them into your Word document. *Figures should use a light background (e.g., do not copy plots while your computer is in dark mode).* All figures must include figure numbers and descriptive captions. Finally, verify figure quality in the PDF version required for submission. (**3 pages including the figures**)

4. Comparison

Compare your results with the theoretical results. Add figures in which you plot your experimental results on top of the results predicted by the theory. Discuss how you have found the theoretical results. The equations for these theoretical results must have been given in Section 1. Here, you say how you solve those equations and find the results. List the parameters that you have used and other steps that one has to consider to come up with the theoretical results. (**1 page**)

5. Discussion

Discuss why your theoretical and experimental results do not match (or do match!). Was what you observe expected or not? If not, what do you think caused the deviation from the theory? If you do not have a simple theoretical model for the system, discuss the general expected behavior, and compare that with what you have observed. Is the deviation because of the experimental error? Was there a mistake in conducting experiments? Is the difference between theoretical and experimental data is within the range that is acceptable? How do you decide on this acceptable deviation from theory? (**1 page**)

6. Conclusions

Summarize what you have achieved in these series of experiments. Very briefly mention the theory, and discuss the differences between your experimental results and theoretical results. What was the take home message? What would you do differently if you were to do these experiments again? What was the main lesson you learned? (**1 page**)

References

Please list the references used in your text. You should not submit a document without any references. At a minimum, you must cite the lecture notes. (**no page limit**)

Appendices

Include plots and tables used in your analysis that were not placed in the main report. An appendix is optional and should only be added when needed. Do not assume the appendix will be read; use it only to support results or claims in the report that require additional data or background to build reader confidence. (**no page limit**)